Worksheet 50 · Sequences – Nth Term IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Continue sequences and find missing terms • Find the nth term of a linear sequence • Decide whether a number is a term of a sequence		
Grade: / 62	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes

n^{th} term → Difference × n + zeroth term

Questions *Sequences – Nth Term*

Question 1 (2 marks)

Here are the first four terms of an arithmetic sequence.

6 10 14 18

(a) Find an expression, in terms of n, for the nth term of this sequence.

(2)

(b) Write down an expression, in terms of n, for the $(n + 1)$th term of this sequence.

Question 2 (2 marks)

Here are the first five terms of an arithmetic sequence.

7 11 15 19 23

Write down an expression, in terms of n, for the nth term of this sequence.

Question 3 (4 marks)

Here are the first five terms of a number sequence S.

10 16 22 28 34

(a) Find an expression, in terms of n, for the nth term of this sequence.

(2)

The nth term of a sequence T is given by $n^2 - 3$

There are numbers that are terms in both the sequence S and the sequence T.

(b) Find one of these numbers.

(2)

Question 4 (5 marks)

Here are the first four terms of a sequence of fractions.

$\frac{1}{1}$ $\frac{2}{3}$ $\frac{3}{5}$ $\frac{4}{7}$

The numerators of the fractions form the sequence of whole numbers 1 2 3 4 ...
The denominators of the fractions form the sequence of odd numbers 1 3 5 7 ...

(a) Write down an expression, in terms of n, for the nth term of this sequence of fractions.

(2)

(b) Using algebra, prove that when the square of any odd number is divided by 4 the remainder is 1

(3)

Question 5 (2 marks)

Here are the first five terms of an arithmetic sequence.

7 10 13 16 19

Find the sum of the first 100 terms of this sequence.

Question 6 (2 marks)

Here are the first 4 terms of an arithmetic sequence.

85 79 73 67

Find an expression, in terms of n, for the nth term of the sequence.

Question 7 (5 marks)

Here are the first five terms of a number sequence.

3 7 11 15 19

(a) Write down the next term of the sequence.

.....

(1)

(b) Explain how you worked out your answer.

.....

.....

(1)

(c) Find the first number greater than 70 that is in the sequence.

.....

(2)

Ada says,

"96 is a number in the sequence"

(d) Is Ada correct?
You must give a reason for your answer.

.....

.....

(1)

Question 8 (4 marks)

Here are the first five terms of a number sequence.

7 11 15 19 23

(a) Find an expression, in terms of n, for the nth term of this sequence.

.....

(2)

The nth term of a different number sequence is given by $80 - 2n$

(b) Write down the first 3 terms of this sequence.

....., ..., ...

(2)

Yuen says there are no numbers that are in both of the sequences.
Yuen is correct.

(c) Explain why.

.....

.....

(1)

Question 9 (32 marks)

Here are the first five terms of an arithmetic sequence.

1 5 9 13 17

(a) Find an expression, in terms of n, for the nth term of this sequence.

.....

(2)

The nth term of another arithmetic sequence is $3n + 5$

(b) Find an expression, in terms of m, for the $(2m)$th term of this sequence.

.....

(1)

Question 10 (2 marks)

The first four terms of an arithmetic sequence are

2 9 16 23

Write down an expression, in terms of n, for the nth term.

Question 11 (2 marks)

Here are the first four terms of an arithmetic sequence.

38 31 24 17

Find an expression, in terms of n, for the nth term of the sequence.

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